ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

TECHNOLOGY

PRACTICE EXAM 3 · MARKING GUIDELINE

MARKS: 75

Marking guideline for the Academic Central Grade 8 Technology practice exam 3.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words and other correct examples. In calculations give method marks even if the final answer is wrong.

QUESTION 1

1 1.1 B ✓ 1.2 D ✓ 1.3 A ✓ 1.4 C ✓ 1.5 C ✓ 1.6 A ✓ 1.7 D ✓ 1.8 B ✓ 1.9 D ✓ 1.10 B ✓ [10]

QUESTION 2

2.1.1 Any suitable type with a reason, e.g. a truss bridge – strong and light, can span 12 m with simple materials ✓✓; a suspension footbridge – needs few supports in the stream; a beam bridge with a central pillar (2)

2.1.2 Any TWO with reasons: steel – strong ✓; treated wood – cheap and easy to work ✓; concrete for the supports – strong in compression and resists water; steel cable – strong in tension (2)

2.1.3 Any TWO: compression ✓; tension ✓; bending; shear; torsion (from wind) (2)

2.1.4 Any TWO: deep, strong foundations ✓; anchor the bridge firmly to the banks ✓; use cross-bracing; build it high above the flood level; use a wide base (2)

2.1.5 Any TWO: handrails ✓; non-slip deck surface ✓; side netting or barriers; a maximum load sign; good lighting (2)

2.2 A strut is in compression (it is pushed or squashed) ✓; a tie is in tension (it is pulled or stretched) ✓. (2)

2.3 Neat sketch ✓; deck shown and labelled ✓; supports shown and labelled ✓; framework made of triangles ✓ (4)

2.4 The material can bend, crack, buckle or break ✓ and the structure can collapse (fail) ✓. (2)

[18]

QUESTION 3

3.1 (a) Cool-drink and water bottles ✓ (b) Milk bottles, crates, shopping bags ✓ (c) Water pipes, electrical insulation, raincoats ✓ (d) Takeaway containers, packaging, cups ✓ (4)

3.2 They do not melt or soften when they get hot ✓ and they are good electrical (and heat) insulators ✓. (2)

3.3 To encourage people to use fewer plastic bags (reuse bags) and so reduce plastic waste ✓ (1)

3.4 Any TWO: use refillable water bottles ✓; bring lunch in reusable containers ✓; recycle plastic in separate bins; avoid single-use straws and bags; reuse plastic for projects (2)

3.5 A plastic made from plant materials (e.g. maize starch) that can break down more easily ✓ (1)

3.6 A mixture of a metal with one or more other elements to improve its properties ✓. Any ONE: steel, brass, bronze, stainless steel, solder ✓ (2)

[12]

QUESTION 4

4.1.1 $48 \div 16 \checkmark = 3 \text{ times} \checkmark$ (2)

4.1.2 $60 \times 3 \checkmark = 180 \text{ r/min} \checkmark$ (2)

4.1.3 $\text{Distance} = 180 \times 2 \text{ m} = 360 \text{ m per minute} \checkmark; 360 \text{ m} \times 60 = 21\,600 \text{ m per hour} \checkmark = 21{,}6 \text{ km/h} \checkmark$ (3)

4.1.4 A high gear gives more speed for each turn of the pedals ✓; on a flat road little force is needed, so the cyclist can go faster ✓. (2)

4.2 The driver gear is turned by the input (motor or hand) ✓; the driven gear is turned by the driver and gives the output ✓. (2)

4.3 $\text{Effort} = 600 \div 3 \checkmark = 200 \text{ N} \checkmark$ (2)

4.4 Belt drive, any ONE: quiet, cheap, absorbs shocks, can slip to protect the machine ✓. Gear drive, any ONE: no slipping, transmits large forces, compact ✓ (2)

4.5 (a) Open belt with both pulleys turning in the same direction ✓ (b) Crossed belt with the pulleys turning in opposite directions ✓ (2)

4.6 A turning force (force that causes rotation) ✓ (1)

[18]

QUESTION 5

5.1.1 AND logic ✓ (1)

Timer	Soil dry	Pump
0	0	0
0	1	0
1	0	0
1	1	1

5.1.2 All four input combinations ✓; output column correct ✓✓✓ (4)

5.1.3 Correct battery symbol ✓; two switches in series ✓; motor symbol (circle with M) ✓; complete closed loop ✓ (4)

5.2 Any TWO with what they detect: LDR – light ✓; thermistor – temperature ✓; push switch / pressure pad – pressure; microphone – sound; moisture sensor – water (2)

5.3 (a) The bulbs become dimmer ✓ (b) The bulbs stay (about) as bright as one bulb ✓ (2)

5.4 $\text{Saved power} = 60 - 9 = 51 \text{ W} \checkmark; \text{energy} = 51 \text{ W} \times 5 \text{ h} = 255 \text{ Wh} \checkmark = 0{,}255 \text{ kWh} \checkmark$ (3)

5.5 An electric motor ✓ (1)

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TOTAL: 75

END OF MARKING GUIDELINE